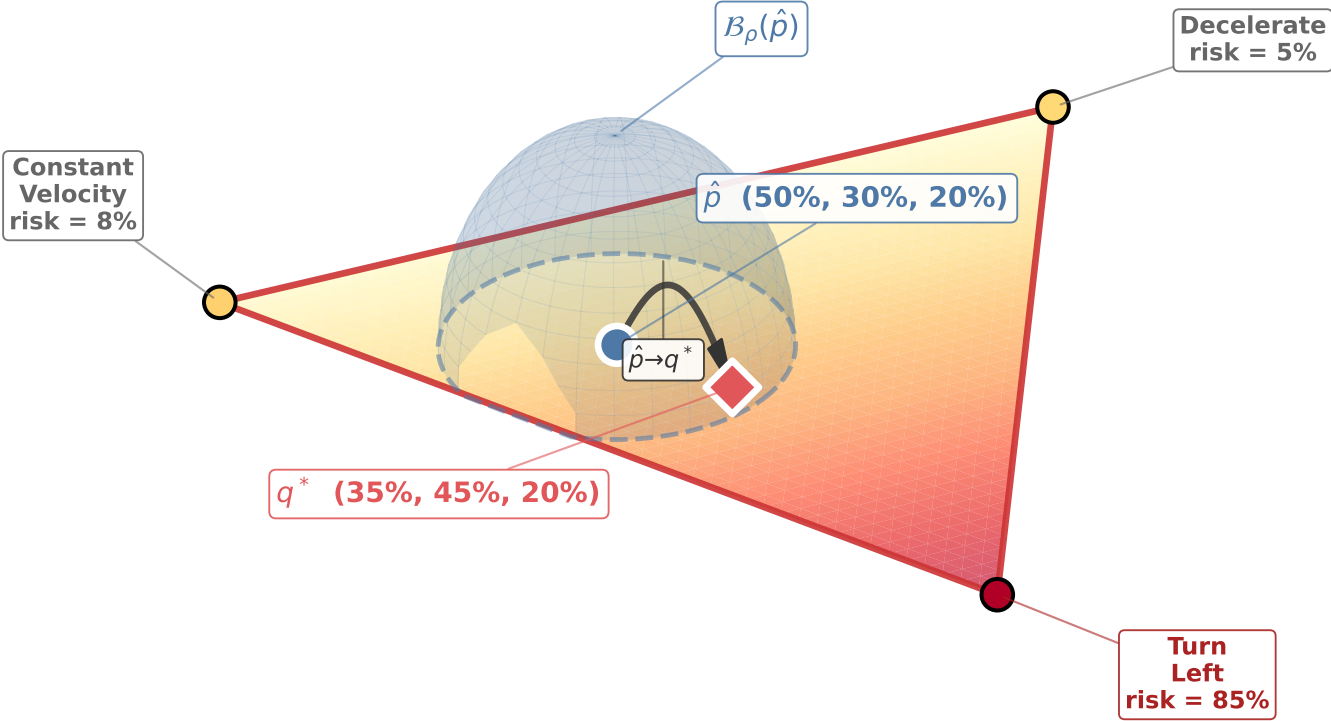


Wasserstein DRO Reweighting on the Probability Simplex



Objective

$$\sup_{q \in \mathcal{B}_\rho(\hat{p})} \sum_m q_m r_m$$

Symbols

q_m	Weight on mode m in candidate distribution q
r_m	Collision risk of mode m (color intensity on simplex)
$\hat{p}$	Nominal distribution (observed mode frequencies)
q^*	Worst-case distribution (maximizes expected risk)
$\mathcal{B}_\rho(\hat{p})$	Wasserstein ball of radius ρ around $\hat{p}$ (ambiguity set)
ρ	Wasserstein radius (controls robustness level)

Why a simplex?

Each point on the simplex is a probability distribution over modes ($q_m \geq 0, \sum_m q_m = 1$).

DRO finds the distribution q^* within the Wasserstein ball that maximizes expected collision risk $\sum_m q_m r_m$, shifting mass toward dangerous modes.